Dataset Provenance, Institutional Origins, and Legal Licensing Protocols

A Formal Audit of PlantVillage, Roboflow Universe (personal-enkcf), and PlantDoc Repositories

1. Executive Overview and Data Governance Framework

The development and deployment of lightweight deep learning architectures for precision phytopathology require strict data governance, institutional provenance tracking, and legal compliance. In computer vision research for agricultural disease diagnosis, training models without transparent data provenance introduces legal liability regarding copyright infringement, severely impairs scientific reproducibility, and obscures underlying dataset biases. When building edge-deployable models such as MobileNetV2 for field use, understanding the origin, collection protocols, and intellectual property terms of target datasets is essential to ensure lawful reuse and robust generalization.

This document presents a comprehensive audit of the three primary computer vision repositories acquired and evaluated within this project: **PlantVillage**, **Roboflow Universe (personal-enkcf)**, and **PlantDoc**. Each dataset represents a distinct stage in the evolution of agricultural AI benchmarks, ranging from controlled laboratory single-leaf photography to handheld smartphone and drone field imagery.

Dataset Name	Institutional Origin & Authors	Publication & Host	License Type	Primary Imaging Modality	Target Crop Scope
PlantVillage	Penn State & EPFL (Hughes & Salathé, 2015)	arXiv:1511.08060 / GitHub / Kaggle	CC-BY 4.0	Controlled Lab (Excised Single Leaves)	14 Crop Species (18,160 Tomato)
Roboflow Universe (personal-enkcf)	Univ. of Naples Parthenope & CECOS Univ. (Islam et al., 2026)	MDPI Automation / Roboflow Universe	Open Access / Public Repo	Handheld Smartphone & Drone Field Imagery	100% Tomato (10,594 Field Images)
PlantDoc	IIT Delhi & IIIT Delhi (Singh et al., 2020)	ACM IKDD CoDS / COMAD 2020	Open Academic Research	Web-Scraped & Unconstrained Field	13 Crop Species (749 Tomato Images)

2. PlantVillage Benchmark Repository

2.1 Institutional Provenance and Author Attribution

The PlantVillage dataset was established in 2015 as a collaborative open-access initiative led by David P. Hughes at Pennsylvania State University and Marcel Salathé at the École Polytechnique Fédérale de Lausanne (EPFL). The primary objective of the initiative was to create a centralized, expert-curated image repository to train automated plant disease diagnostic algorithms. The repository is formally documented in the foundational paper titled *"An open access repository of images on plant health to enable the development of mobile disease diagnostics through machine learning and crowdsourcing"* published on arXiv (arXiv:1511.08060).

2.2 Legal Licensing Terms and Usage Rights

PlantVillage is distributed under the **Creative Commons Attribution 4.0 International License (CC-BY 4.0)**. Under the terms of CC-BY 4.0, researchers and commercial developers are granted the rights to copy, redistribute, transform, and build upon the material for any purpose, including commercial applications. The sole legal requirement is providing appropriate credit to the original authors (Hughes & Salathé, 2015), linking to the CC-BY 4.0 license, and indicating if modifications were made to the original dataset.

2.3 Acquisition Protocols and Image Characteristics

The plant leaves in the PlantVillage repository were collected from research greenhouses and field plots, after which single leaf specimens were excised from the host plants. Imaging was conducted indoors under standardized artificial lighting. Leaves were flattened and positioned against uniform, monochrome background sheets consisting of gray, white, or light blue poster board. The tomato subset contains 18,160 images spanning 10 distinct diagnostic classes (9 disease categories and 1 healthy control).

2.4 Methodological Vulnerabilities and Shortcut Learning Risks

Extensive literature in agricultural computer vision demonstrates that models trained exclusively on PlantVillage suffer from severe **shortcut learning**. Because every leaf is excised and presented on a clean monochrome background, convolutional neural networks

frequently memorize background color temperature, lighting angles, and leaf boundary geometry rather than genuine pathological lesions (such as chlorotic halos or concentric necrotic rings). Consequently, models that achieve >99% validation accuracy on PlantVillage collapse when deployed on real-world field images with natural soil and canopy clutter. Additionally, PlantVillage contains sequential rotation shots of identical physical leaves, creating within-dataset burst-shot leakage if splits are performed randomly.

3. Roboflow Universe Open-Field Dataset (personal-enkcf)

3.1 Repository Provenance and Institutional Authorship

The Roboflow Universe Tomato Leaf Disease Dataset was curated and hosted under the public repository account personal-enkcf (tomato-leaf-disease-detection-gyozv). The dataset and its application in real-time disease detection and automated precision spray prescription were formally validated and published by Shahab Ul Islam, Giampaolo Ferraioli, Ghassan Husnain, Abdul Waheed, and Vito Pascazio (2026) in the MDPI journal *Automation* (Volume 7, Issue 2, Article 36), representing a joint research collaboration between the Department of Engineering at the University of Naples Parthenope, Italy, and CECOS University of IT and Emerging Sciences, Pakistan.

3.2 Licensing and Public Repository Access

The dataset is hosted on Roboflow Universe as a public open-access computer vision dataset. Re-use for academic research and model benchmarking is governed by Roboflow's public data sharing terms alongside the open-access publication terms of MDPI (CC-BY 4.0). Users accessing the dataset programmatically via the Roboflow API or raw downloads are required to maintain proper academic attribution to Shahab Ul Islam et al. (2026).

3.3 Field Acquisition Modalities and Spatial Coverage

Unlike laboratory baselines, 100% of the 10,594 images in the Roboflow dataset represent real-world field conditions. The imaging methodology combined handheld smartphone photography with low-altitude unmanned aerial vehicle (UAV) drone flights across tomato plots. The captured images incorporate dynamic outdoor sunlight, direct canopy shadows, background soil textures, weed intrusions, and irrigation drip lines across multiple foliage growth stages. Annotations include both image-level disease labels and bounding-box spatial coordinates compatible with single-stage object detectors (such as YOLOv8 and YOLOv9).

3.4 Class Imbalance and Data Leakage Considerations

Empirical audit of the Roboflow dataset reveals significant class imbalance across its 9 categories: Leaf Mold (4,532), Early Blight (3,000), Leaf Miner (1,234), Mosaic Virus (1,234), Late Blight (1,200), Septoria Leaf Spot (432), Healthy (432), Yellow Leaf Curl Virus (124), and Spider Mites (43). To prevent model bias toward majority classes, training pipelines must employ weighted Cross-Entropy loss or Focal Loss. Furthermore, because drone footage captures consecutive video frames milliseconds apart, strict campaign-level partitioning is necessary to prevent sequential video frame leakage between training and validation sets.

4. PlantDoc Open-Field and Web-Scraped Benchmark

4.1 Institutional Origins and Primary Literature

PlantDoc was developed in 2020 by researchers at the Indian Institute of Technology (IIT) Delhi and the Indraprastha Institute of Information Technology (IIIT) Delhi, authored by Devyam Singh, Naman Jain, Pranjali Jain, Pratik Kayal, Shrey Kumawat, and Nitesh Batra. The benchmark was presented at the 7th ACM IKDD CoDS and 25th COMAD conference (ACM, 2020, pp. 249–253) under the title *"PlantDoc: A Dataset for Visual Plant Disease Detection"*.

4.2 Legal Rights and Academic Research Licensing

PlantDoc is made available as an open-access academic research dataset hosted via GitHub (pratikkayal/PlantDoc-Dataset). The dataset is intended for non-commercial academic research, computer vision benchmarking, and educational evaluation. Publications utilizing the dataset are required to cite the original ACM conference proceedings paper by Singh et al. (2020).

4.3 Data Curation, Web-Scraping Protocols, and Annotations

PlantDoc was created to address the critical gap between laboratory trained models and field reality. The dataset authors spent 352 human-hours scraping images from public domain web sources, search engines, agricultural blogs, and field photography, followed by manual bounding-box annotation. The complete dataset comprises 3,451 images across 13 crop species and 27 categories, with the tomato subset accounting for 749 images spanning 7 folder categories.

4.4 Technical Limitations and Categorical Gaps

Because PlantDoc was assembled via web scraping: (1) **Resolution Heterogeneity**: Images exhibit extreme variance in spatial resolution, compression artifacts, and watermarks. (2) **Generic Class Aggregation**: Viral pathologies involving foliage leaf curl are aggregated under a single broad folder label (Tomato Leaf Curl), requiring taxonomy harmonization. (3) **Complex Background Noise**: Images feature non-target foliage, background weeds, and hand-held leaf holding, making full-image classification challenging without localization pre-processing.

5. Legal Governance, Commercial Compliance, and Deployment Roadmap

5.1 Commercial vs. Academic Licensing Matrix

When transitioning deep learning models from academic research to field deployment on mobile devices or IoT edge microcontrollers, licensing constraints dictate how datasets can be utilized across the model development lifecycle.

Dataset Repository	License Type	Academic Research Permitted	Commercial Model Training Permitted	Required Attribution Citation
PlantVillage	CC-BY 4.0	Yes	Yes	Hughes & Salathé (2015), arXiv:1511.08060
Roboflow Universe (personal-enkcf)	Open Access / Public Terms	Yes	Yes (Standard Open Terms)	Shahab Ul Islam et al. (2026), MDPI Automation
PlantDoc	Open Academic Research	Yes	Conditional (Research Re-use)	Singh et al. (2020), ACM CoDS/COMAD

5.2 Data Provenance and Governance Protocol

- **Attribution Integrity**: Maintain formal citations in code documentation, model cards, and published literature for all three dataset sources.
- **Data Leakage Isolation**: Enforce absolute out-of-distribution (OOD) isolation for **PlantDoc**, preserving it exclusively as a zero-shot validation benchmark without exposing its images to training or hyperparameter tuning loops.
- **Transparent Model Cards**: Document exact dataset split sizes, class mappings, pre-processing transformations, and training loss functions in model deployment specifications.

> ***Summary Compliance Note:*** *All three active datasets (PlantVillage, Roboflow Universe personal-enkcf, and PlantDoc) permit academic research and development. Both PlantVillage and Roboflow dataset terms permit commercial model training provided proper attribution is maintained. PlantDoc remains strictly isolated as an uncoraminated zero-shot cross-domain evaluation set.*